

Display unit failures and A380X reconfiguration

flyPad failures for the A32NX ND and ECAM display units and the eight A380X CDS display units: the failed display goes blank, with the A380X CDS ... DU FAULT alert; the A380X moves the PFD and the EWD to another display unit, with the PFD/ND and DU RECONF pushbuttons; the DU RECONF pb also shows the EWD and the SD on the ND display unit

Aircraft	A32NX and A380X
Where to find it	flyPad > Failures > ATA 31 Indicating / Recording System; the display units; A380X E/WD and SD STATUS; A380X PFD/ND and DU RECONF pushbuttons
Branch	core/display-unit-failures , based on core/nd-weather-terrain (stacked: needs core/nd-weather-terrain merged first, the radar / terrain gauge goes blank with a failed ND).
Status	All three features tested in the sim by the user: OK (2026-10-04), the EWD and SD on the ND display unit included (load, CAPT, F/O, both sides).
Pictures	No pictures: a failed display unit is blank, the alerts are the stock FBW E/WD and SD displays.

1. A32NX ND and ECAM display unit failures

Aircraft:	A32NX		Where:	flyPad > Failures > ATA 31; CPT / F/O ND, upper and lower ECAM		Status:	Tested in the sim by the user: OK (2026-10-04).
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What it is for

The flyPad could fail the two PFDs only. Four more display units (DUs) can now be failed: the CAPT and F/O NDs and the upper and lower ECAM DUs. A failed DU goes blank, the way a failed PFD already did. The A320 FCOM (DSC-31-05-60, FAILURE OF A DU) lists a blank screen with an amber F, a distorted display or a blank screen with INVALID DISPLAY UNIT; the blank (OFF) screen is used here.

- The weather radar and terrain picture of the ND (the native gauge over it) goes blank with a failed ND.
- It is also no longer drawn on an ND that is switched off (brightness knob OFF) or still in its self-test.
- The lower ECAM failure blanks the whole SD (both of its display layers). Before, the SD display unit ignored a failure change.
- When the failure is cleared, the DU runs its self-test, then shows its picture again.

The flyPad failures (ATA 31 Indicating / Recording System)

flyPad failure name	Display unit that goes blank
Captain PFD display	CAPT PFD (existing failure)
F/O PFD display	F/O PFD (existing failure)
Captain ND display	CAPT ND, with its weather / terrain picture
F/O ND display	F/O ND, with its weather / terrain picture
Upper ECAM display	Upper ECAM DU (E/WD)
Lower ECAM display	Lower ECAM DU (SD)

Limitations and why

Limitation	Why	What to do
No ECAM alert for a failed DU.	The branch only blanks the DU.	-

No automatic transfer of a picture to another DU.	The FCOM transfers (E/WD to the lower ECAM DU, PFD to the ND DU, DSC-31-05-60) are not part of this branch.	Use the ECAM / EIS switching as available.
Always a blank screen.	The amber F, the distorted picture and INVALID DISPLAY UNIT of the FCOM are not drawn.	-

How to use

1. Open **flyPad > Failures**, chapter **ATA 31 Indicating / Recording System**.
2. Activate one of the display failures (table above): that DU goes blank.
3. Deactivate it to get the DU back (after its self-test).

How to test

#	Do this	Expected result
1	A32NX, any phase, all displays on. CPT ND in ARC with the weather radar on (weather around) or TERR ON ND. flyPad > Failures > ATA 31: activate Captain ND display .	The CPT ND goes blank, with no weather or terrain left on it. The F/O ND is not affected.
2	Deactivate Captain ND display .	The CPT ND shows its self-test, then the ND picture with the weather / terrain.
3	Activate F/O ND display , then deactivate it.	Same on the F/O ND.
4	Activate Upper ECAM display .	The upper ECAM DU (E/WD) goes blank.
5	Deactivate it; activate Lower ECAM display .	The E/WD comes back after its self-test; the lower ECAM DU (SD) goes blank, whatever SD page was shown.
6	Deactivate it.	The SD comes back after its self-test.
7	No failure. With the weather radar picture on the CPT ND, turn the CPT ND brightness knob fully OFF, then back on.	The ND and its weather picture go dark together (no weather left on the dark ND) and come back together.

2. A380X CDS display unit failures

Aircraft: A380X | **Where:** flyPad > Failures > ATA 31; the 8 CDS DUs; E/WD; SD STATUS | **Status:** Tested in the sim by the user: OK (2026-10-04).

What it is for

The eight display units of the A380 CDS (A380 FCOM DSC-31-15-10: CAPT PFD, CAPT ND, CAPT MFD, E/WD, SD, F/O MFD, F/O ND, F/O PFD) can be failed from the flyPad. The failed DU goes blank (on an ND, the weather radar and terrain picture too, the VD included) and the FWS raises the **CDS ... DU FAULT** alert with its FCOM procedure and its INOP SYS line (FCOM PRO-ABN-ECAM-10-31).

The flyPad failures (ATA 31 Indicating / Recording System)

flyPad failure name	ECAM alert (amber)	STATUS INOP SYS
CAPT PFD display unit	CDS CAPT PFD DU FAULT	CAPT PFD DU
CAPT ND display unit	CDS CAPT ND DU FAULT	CAPT ND DU
CAPT MFD display unit	CDS CAPT MFD DU FAULT	CAPT MFD DU
EWD display unit	CDS EWD DU FAULT	EWD DU

SD display unit	CDS SD DU FAULT	(none: the FCOM lists no SD line)
F/O MFD display unit	CDS F/O MFD DU FAULT	F/O MFD DU
F/O ND display unit	CDS F/O ND DU FAULT	F/O ND DU
F/O PFD display unit	CDS F/O PFD DU FAULT	F/O PFD DU

What you should see

Situation	What you should see
Failure activated	The DU goes blank. Single chime, MASTER CAUT, and the amber alert on the E/WD.
ECAM procedure (each DU)	xx DU OFF 5 S THEN ON; IF NOT SUCCESSFUL: xx DU OFF; DU RECONF P/B AVAIL (green). The SD DU FAULT adds: IF SD DU OFF : F/O DU RECONF P/B FOR ATC MAILBOX.
SD STATUS page	INOP SYS (all phases): the DU line of the table above.
Takeoff and landing	The alert is inhibited in the FWS flight phases 3 to 7, 9 and 10 (FCOM PDF p. 5386); the DU is blank anyway.
DU without power	No DU FAULT alert (design choice): the lost busbar has its own ELEC alert.
Failure cleared	The DU runs its self-test and shows its picture again; the alert and the INOP SYS line go.

Limitations and why

Limitation	Why	What to do
The reconfiguration has limits.	See "A380X display unit reconfiguration" below: the PFD and the EWD move to another DU by themselves, the MFD never does, and the SD only on an ND DU with the DU RECONF pb, without the ATC mailbox.	With the SD DU failed, press the F/O DU RECONF pb (it acts on the F/O ND DU) until the SD pages (STATUS included) show there.
The OFF 5 S THEN ON item does not bring the DU back.	The flyPad failure stays until it is cleared, which is the procedure's NOT SUCCESSFUL branch.	Clear the failure in the flyPad.
The second FCOM trigger (a DU that declared two other DUs faulty in turn) is not modelled.	It needs the DU monitoring loops, which are not simulated.	-

How to use

1. Open **flyPad > Failures**, chapter **ATA 31 Indicating / Recording System**.
2. Activate one of the eight ... **display unit** failures.
3. Follow the ECAM procedure, then check the SD STATUS page.
4. Deactivate the failure to get the DU back.

How to test

#	Do this	Expected result
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1	A380X on the ground (before the takeoff power) or in cruise, all DUs on. CPT ND in ARC with the WX overlay and weather around. flyPad > Failures > ATA 31: activate CAPT ND display unit .	The CAPT ND and its VD go blank, with no weather or terrain left. Single chime, MASTER CAUT. E/WD: CDS CAPT ND DU FAULT with CAPT ND DU OFF 5 S THEN ON, IF NOT SUCCESSFUL, CAPT ND DU OFF, DU RECONF P/B AVAIL in green.
2	Open the SD STATUS page.	INOP SYS: CAPT ND DU .
3	Deactivate the failure.	The CAPT ND runs its self-test, then shows the ND, the VD and the weather again; the alert and the INOP SYS line are gone.
4	Repeat with CAPT PFD display unit, CAPT MFD display unit, F/O PFD display unit, F/O ND display unit and F/O MFD display unit .	Each time the named DU goes blank and the E/WD shows the matching CDS ... DU FAULT, with its INOP SYS line on the STATUS.
5	Activate SD display unit .	The SD goes blank; E/WD: CDS SD DU FAULT with the extra green line IF SD DU OFF : F/O DU RECONF P/B FOR ATC MAILBOX (the FCOM gives this alert no INOP SYS line).
6	Deactivate it; activate EWD display unit .	The EWD DU goes blank and the E/WD moves to the SD DU (automatic reconfiguration), where CDS EWD DU FAULT can be read; single chime and MASTER CAUT; INOP SYS EWD DU on the STATUS.
7	Activate any DU failure during the takeoff roll (above takeoff power).	The DU goes blank but no alert until the end of the inhibition.

3. A380X display unit reconfiguration

Aircraft: A380X | **Where:** CAPT and F/O PFD/ND and DU RECONF pushbuttons; the 8 CDS DUs | **Status:** Tested in the sim by the user on all steps: OK (2026-10-04). Unit tests: CdsReconfiguration.spec.ts (shared and systems host), HostedDisplay.spec.ts, hostedInstrument.spec.ts; native ndwxr test cds_display_test.

What it is for

The A380 CDS moves a lost display to another display unit (A380 FCOM DSC-31-15-20 CDS abnormal operations, DSC-31-20-50 PFD/ND pb). Before, a failed or switched-off DU simply lost its picture. Now:

- **Automatic:** when a PFD DU fails, loses its power or is switched off, the PFD is shown on the ND DU of the same side (the ND is lost). When the EWD DU is lost, the E/WD is shown on the SD DU.
- **PFD/ND pushbutton:** exchanges the on-side PFD and ND. With the exchange selected, the PFD is on the ND DU and comes back to the PFD DU if the ND DU fails (the PFD keeps the priority).
- **DU RECONF pushbutton:** active only with a failed on-side DU; it acts on the DU the FCOM gives (CAPT: the MFD DU, then the ND DU if the MFD DU is failed, then the PFD DU if the ND and MFD DUs are failed; F/O: the ND DU, the MFD DU if the PFD or ND DU is failed, then the PFD DU) and steps through the displays of the FCOM order that this DU can draw, then back to its normal display.
- **EWD and SD on the ND DU:** with the CAPT (F/O) PFD and MFD DUs failed, the pb acts on the ND DU, which shows the PFD automatically; it now cycles **PFD > EWD > SD > PFD** (the MFD of the FCOM sequence cannot be drawn there). With only the MFD DU (CAPT) or the SD DU (F/O) failed, the ND DU cycles ND > EWD > SD > ND the same way. The EWD (SD) is never shown on both sides at once (A380 FCOM DSC-31-15-20), and a PFD that moves to the ND DU later (its PFD DU fails afterwards) wins over a selection there.
- The weather radar and terrain picture (ndwxr) is hidden while its ND DU shows another display (the PFD, the EWD or the SD).

What you should see

Situation	What you should see
CAPT PFD display unit failed (or switched off)	The CAPT PFD DU is blank and the PFD appears on the CAPT ND DU, without weather or terrain on it.
EWD display unit failed	The E/WD appears on the SD DU.
PFD/ND pb pressed (CAPT or F/O)	The PFD and the ND of that side swap DUs; press again to swap back.
CAPT ND and MFD DUs failed, DU RECONF pressed	The CAPT PFD DU steps from the PFD to the ND (with no weather or terrain image), then back to the PFD.
CAPT PFD and MFD DUs failed, DU RECONF pressed	The CAPT ND DU, which shows the PFD, steps to the E/WD, then to the SD, then back to the PFD. While the E/WD (SD) is shown there, the F/O pb skips it.
SD DU failed, F/O DU RECONF pressed	The F/O ND DU steps from the ND to the E/WD, then to the SD pages (without the ATC mailbox), then back to the ND.
Failure cleared	Each DU goes back to its normal display.

Limitations and why

Limitation	Why	What to do
The MFD is never drawn on another DU, and nothing is drawn on the MFD DUs; the DU RECONF pb skips them, so it does nothing when it acts on an MFD DU.	Every display is a gauge of its own on its DU texture: the MFD is one gauge over both MFD screens (one texture) that also runs the FMS. Design choice: the PFD and the ND on both DUs of a side, the E/WD and the SD also on the ND DUs, and the E/WD on the SD DU.	Use the other side's MFD.
The SD on an ND DU has no ATC mailbox.	The mailbox stays with the SD DU's own SD (design choice of the hosted SD).	With the SD DU failed, the ATC mailbox is lost.
A selected E/WD or SD replaces the PFD on that side.	With the PFD and MFD DUs of a side failed, the ND DU is the only DU left for the PFD (FCOM: the pb acts on the ND DU).	Press DU RECONF again to get the PFD back.
The ND on the PFD DU has no weather radar or terrain image.	The native ndwxr gauges stay on the ND DUs.	Use the other side's ND.
The KCCU keys do not reconfigure the DUs.	Not modelled.	Use the DU RECONF pb.
A full MSFS restart is needed after an update.	panel.cfg (new hosted gauges, the E/WD and the SD on the ND DUs included) and the cockpit behaviour XML change.	Restart the sim after installing.

How to use

1. Nothing to do for the automatic reconfiguration: a lost PFD or EWD DU moves its display by itself.
2. Press the **PFD/ND** pushbutton of your side to exchange your PFD and ND.
3. With a failed on-side DU, press **DU RECONF** to step the display of the DU it acts on.
4. With your PFD and MFD DUs failed, press **DU RECONF** to bring the E/WD, then the SD, onto your ND DU; once more for the PFD.

How to test

#	Do this	Expected result
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1	A380X powered, all DUs on. flyPad > Failures > ATA 31: activate CAPT PFD display unit .	The CAPT PFD DU goes blank; the PFD appears on the CAPT ND DU (no weather / terrain image on it); CDS CAPT PFD DU FAULT.
2	Deactivate it. Press the CAPT PFD/ND pushbutton.	The CAPT PFD and ND swap DUs. Press again: back to normal.
3	Turn the F/O PFD DU brightness knob OFF.	The PFD appears on the F/O ND DU. Turn it back on: normal again.
4	Activate EWD display unit .	The E/WD appears on the SD DU. Deactivate it: the SD comes back.
5	Activate CAPT ND display unit and CAPT MFD display unit , then press the CAPT DU RECONF pushbutton twice.	The CAPT PFD DU shows the ND, then the PFD again.
6	Deactivate them. Activate CAPT PFD display unit and CAPT MFD display unit .	The PFD appears on the CAPT ND DU.
7	Press the CAPT DU RECONF pushbutton three times.	The CAPT ND DU shows the E/WD, then the SD (the current SD page), then the PFD again. No weather or terrain image on it meanwhile.
8	Press the CAPT DU RECONF pb once (E/WD on the CAPT ND DU). Then activate F/O PFD display unit and F/O MFD display unit and press the F/O DU RECONF pb.	The F/O ND DU skips the E/WD (shown on the CAPT side) and shows the SD.
9	Deactivate all failures. Activate SD display unit and press the F/O DU RECONF pb twice.	The F/O ND DU shows the E/WD, then the SD pages (no ATC mailbox); once more: the ND again.
10	Deactivate all failures.	Every DU shows its normal display.